综合练习

人员

于子珈、杨瑾硕、周子一、陈洛冉、司云心、赵熙羽 到课, 李子瀚 线上

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3300

比赛概况 题目列表 选择题列表 提交记录 实时榜单 选择题排行榜 编辑比赛

王向东老师周日十点半C++LCA

刷新

#	用户名	姓名	编程分	时间	A	B	C
1	yuzijia1	于子珈	200	5799	100	100	
2	zhaoxiyu	赵熙羽	200	6777	100	100	
3	yangjinshuo	杨谨硕	195	4185	100	95	
4	chenluoran	陈洛冉	100	4005	100		
5	qinxiansen	秦显森	100	4478	100		
6	siyunxin	司云心	100	4655	100		
7	lizihan	李子瀚	100	5566	100		
8	zhouziyi	周子一	100	6956	100		
9	xieyakai	谢亚锴	100	8050	100	0	

本周作业

https://cppoj.kids123code.com/contest/3409 (课上讲了 A ~ C 题, 课后作业是 C 题)

课堂表现

同学们课上整体做题表现还不错, 就是 B 题有几位同学没有用 ST 表做, 建议这些同学用 ST 表再做一下, 加深一下对 ST 表的掌握。

课堂内容

[蓝桥杯 2022 国 B] 机房

对每个点的 deg 维护一个前缀和, 然后后续用 LCA 就可以 O(1) 求了

```
#include <bits/stdc++.h>

using namespace std;

const int N = 1e5 + 5, M = 20;
vector<int> vec[N];
int dis[N], f[N][M], deg[N], p[N];

void dfs(int u, int fa, int d) {
```

```
  dis[u] = d; f[u][0] = fa; p[u] = p[fa] + deg[u];
  for (int i = 1; i < M; ++i) f[u][i] = f[f[u][i-1]][i-1];
  for (int i : vec[u]) {
    if (i == fa) continue;
    dfs(i, u, d+1);
  }
}

int lca(int a, int b) {
  if (dis[a] < dis[b]) return lca(b, a);

  for (int i = M-1; i >= 0; --i) {
    if (dis[f[a][i]] >= dis[b]) a = f[a][i];
  }

  if (a == b) return a;

  for (int i = M-1; i >= 0; --i) {
    if (f[a][i] != f[b][i]) a = f[a][i], b = f[b][i];
  }
  return f[a][0];
}

int main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n-1; ++i) {
    int a, b; cin >> a >> b; ++deg[a], ++deg[b];
    vec[a].push_back(b), vec[b].push_back(a);
  }

  dfs(1, 0, 1);

  while (m -- ) {
    int a, b; cin >> a >> b;
    if (a == b) cout << deg[a] << endl;
    else {
      int fa = lca(a, b);
      cout << p[a] + p[b] - 2*p[fa] + deg[fa] << endl;
    }
  }
  return 0;
}
```

良好的感觉

枚举最小值 i, 找最左和最右, 跟之前的 [蓝桥杯 2023 国 C] 最大区间 这道题基本一样

```
#include <bits/stdc++.h>

using namespace std;
```

```
typedef long long LL;
const int N = 3e5 + 5, M = 20;
int w[N], f[N][M], _lg2[N];
LL p[N];
int n;

LL get_sum(int l, int r) { return p[r] - p[l-1]; }

int get_min(int l, int r) {
  int len = r - l + 1;
  int k = _lg2[len];
  return min(f[l][k], f[r-(1<<k)+1][k]);
}

int find_L(int id) {
  int l = 1, r = id;
  while (l <= r) {
    int mid = (l + r) / 2;
    if (get_min(mid,id) == w[id]) r = mid-1;
    else l = mid+1;
  }
  return l;
}

int find_R(int id) {
  int l = id, r = n;
  while (l <= r) {
    int mid = (l + r) / 2;
    if (get_min(id,mid) == w[id]) l = mid+1;
    else r = mid-1;
  }
  return r;
}

LL calc(int id) {
  int l = find_L(id), r = find_R(id);
  return (LL)w[id] * get_sum(l,r);
}

int main()
{
  for (int i = 0; (1<<i) < N; ++i) _lg2[1<<i] = i;
  for (int i = 1; i < N; ++i) {
    if (!_lg2[i]) _lg2[i] = _lg2[i-1];
  }

  cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i], f[i][0] = w[i], p[i] = p[i-1] + w[i];
  for (int k = 1; k < M; ++k) {
    for (int i = 1; i+(1<<k)-1 <= n; ++i) {
      f[i][k] = min(f[i][k-1], f[i+(1<<(k-1))][k-1]);
    }
  }
```

```
  LL res = 0;
  for (int i = 1; i <= n; ++i) res = max(res, calc(i));
  cout << res << endl;
  return 0;
}
```

[JRKSJ R1] JFCA

用 ST 表维护区间最大的 a 值

破环成链, 把数组拷贝 3 份, 针对中间这一份数组, 向左或向右进行二分, 找到最近的符合要求的点 j

```
#include <bits/stdc++.h>

using namespace std;

const int N = 1e5 + 5, M = 20;
const int inf = 0x3f3f3f3f;
int a[3*N], b[3*N], f[3*N][M], _lg2[3*N];

int get_max(int l, int r) {
  int k = _lg2[r-l+1];
  return max(f[l][k], f[r-(1<<k)+1][k]);
}

int main()
{
  for (int i = 0; (1<<i) < 3*N; ++i) _lg2[1<<i] = i;
  for (int i = 1; i < 3*N; ++i) {
    if (!_lg2[i]) _lg2[i] = _lg2[i-1];
  }

  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> a[i], a[i+2*n] = a[i+n] = a[i];
  for (int i = 1; i <= n; ++i) cin >> b[i], b[i+2*n] = b[i+n] = b[i];

  for (int i = 1; i <= 3*n; ++i) f[i][0] = a[i];
  for (int k = 1; k < M; ++k) {
    for (int i = 1; i+(1<<k)-1 <= 3*n; ++i) {
      f[i][k] = max(f[i][k-1], f[i+(1<<(k-1))][k-1]);
    }
  }

  for (int i = n+1; i <= n+n; ++i) {
    int lLen = inf, rLen = inf;
    if (get_max(i-n+1,i-1) >= b[i]) {
      int l = i-n+1, r = i-1;
      while (l <= r) {
        int mid = (l + r) / 2;
        if (get_max(mid,i-1) >= b[i]) l = mid+1;
```

```
        else r = mid-1;
      }
      lLen = i - r;
    }

    if (get_max(i+1, i+n-1) >= b[i]) {
      int l = i+1, r = i+n-1;
      while (l <= r) {
        int mid = (l + r) / 2;
        if (get_max(i+1,mid) >= b[i]) r = mid-1;
        else l = mid+1;
      }
      rLen = l - i;
    }

    int res = min(lLen, rLen);
    cout << (res==inf ? -1 : res) << " ";
  }
  cout << endl;
  return 0;
}
```